IFRS 9 - impairment - compliance audit

IFRS 9 Financial Instruments - impairment (5.5 and B5.5): staging, significant increase in credit risk, 12-month and lifetime expected credit losses, forward-looking information, write-off.

Malawi Agricultural and Industrial Investment Company (MAIIC) - IFRS 9 EIR and ECL system - build of 14 October 2026 (branch eir_revenue_recognition, commit abe140d) against specification v4. As of 14 October 2026: the code at commit abe140d and the fresh bootstrap maiic_ifrs9_results2 of 13 October 2026 (made before commit 216aee2, so its discounting and Mega Farm figures predate those changes), read with SELECTs only; the Governance Centre's staging and forward-looking settings; the PHPUnit suite as written (not run for this audit). Main book at September 2026 on that bootstrap: 134 loans, gross 23.19bn, ECL 11.57bn undiscounted, 7.12bn discounted on the 62 loans the time-phased engine resolved (72 unresolved). Reviewer: Dr Thomson Kumwenda (CFO), then Deloitte. The Excel workbook of the same name is the working copy (drop-down statuses, hyperlinked contents, reviewer sign-off column, a live Baselines sheet); this document is its printable twin.

Source: IFRS 9 Financial Instruments as issued by the IASB, section 5.5 and B5.5.1 to B5.5.55; 5.4.4 and B5.4.9 (write-off). The Reserve Bank of Malawi Financial Services (Credit Risk Management for Development Finance Institutions) Directive 2018 (docs/regulatory). Specification: docs/MAIIC_EIR_Engine_Specification_v4_2026-10-07.md sections 3.6, 13, 14, 15, 16.

Generated 2026-10-14 10:44:30; the Baselines were read from the database `maiic_ifrs9_wire`.

Status summary

Status	Meaning	Sections
Done	Implemented in the system and visible in an approved output; the test in column 11 fails if the behaviour changes.	9
Partially done	The mechanics are in place, but a parameter, an input or a piece of evidence still differs from the requirement.	3
Outstanding	Required for MAIIC and not built.	0
Not applicable, documented	Does not apply to MAIIC by a recorded decision or fact (reason and reference given); revisit if the facts change.	0
Evidence needed from MAIIC	The system is ready; MAIIC must supply a document, minute or dataset before the row can be closed.	2
Total		14

Contents

Ref	Section	Status	Governance setting	Test
5.5 RECOGNITION OF EXPECTED CREDIT LOSSES				
5.5.1	A loss allowance for expected credit losses	Done		Tests\Feature\Ecl\EclGoldenNumberTest::it_computes_golden_ecl_per_stage_for_prefli_collection_lgd; Tests\Feature\Ecl\TimePhasedEclServiceTest

Ref	Section	Status	Governance setting	Test
5.5.3	Lifetime ECL on a significant increase in credit risk	Done	dpd_basis	Tests\Feature\Ecl\TimePhasedEclServiceTest::test_a_stage_two_lifetime_compounds_beyond_the_twelve_month_pd; Tests\Feature\Eir\StagingServiceTest::test_the_trigger_reads_the_chart_in_force_at_the_month_end_not_a_later_one
5.5.5	12-month ECL where credit risk has not increased significantly	Done		Tests\Feature\Ecl\TimePhasedEclServiceTest::test_stage_one_builds_a_twelve_month_marginal_pd_curve_and_discounts_each_shortfall
5.5.9 to 5.5.11	Determining a significant increase in credit risk; the 30-days-past-due rebuttable presumption	Done	dpd_basis; staging_rebuttal (recorded, not yet applied)	Tests\Feature\Eir\StagingThresholdSeederTest::test_the_long_term_stage_2_rebuttal_is_inactive_until_signed; Tests\Feature\Eir\GoLiveControlsTest::test_a_threshold_is_proposed_by_one_person_and_approved_by_another_before_staging_reads_it; Tests\Feature\Eir\StagingClassifierTest::test_with_no_rule_in_force_the_classifier_stops_and_names_it; Tests\Feature\Eir\StagingServiceTest
B5.5.37	Definition of default; the 90-days-past-due rebuttable presumption	Done	dpd_basis; stage3_missed_instalments	Tests\Feature\Eir\StagingServiceTest::test_the_directive_thresholds_by_tenor_class_and_the_bucket_fallback; Tests\Feature\Eir\StagingThresholdSeederTest::test_medium_and_long_term_facilities_are_stage_3_at_181_days_with_the_rebuttal_cited
5.5 MEASUREMENT				
5.5.17	Measurement of expected credit losses	Partially done	scenario_weighting_method; fli_adjustment_route	Tests\Feature\FLI\FliRouteServiceTest::test_the_ecl_on_the_post_fli_pd_is_the_weighted_ecl_across_scenarios; Tests\Feature\Ecl\TimePhasedEclServiceTest::test_scenario_weights_combine_into_one_allowance
B5.5.41 to B5.5.43	Probability-weighted outcomes	Done	scenario_minimum_count; scenario_weight_bounds; scenario_calibration_note	Tests\Feature\Scenario\ScenarioSetServiceTest::test_a_set_that_breaks_a_rule_is_refused_with_the_reason

Ref	Section	Status	Governance setting	Test
B5.5.49 to B5.5.54	Reasonable and supportable information; forward-looking information	Done	fli_expected_sign_test; fli_r2_cutoff; fli_min_observations; fli_alpha; macro_source_precedence	Tests\Feature\FLI\FliBridgeServiceTest; Tests\Feature\Macro\MacroImportPrecedenceTest
B5.5.52 (transmission)	How forward-looking information reaches the PD	Done	fli_transmission_method; fli_asset_correlation	Tests\Feature\FLI\TransmissionMethodCatalogueTest
B5.5.1 to B5.5.6, B5.5.28 to B5.5.30	Collective assessment and segmentation	Done	pd_segmentation_basis; pd_segment_min_observations; pd_segment_thin_rule; lgd_segment_min_cohort	Tests\Feature\Pd\PdSegmentationTest::test_each_portfolio_gets_its_own_pd_or_its_parents_with_the_lineage_recorded; Tests\Feature\Ecl\MegaFarmProgrammeSeparationTest::test_the_programme_portfolio_is_kept_out_of_the_maiic_scope_and_a_blended_book_is_refused
B5.5.55	Loss given default and collateral	Partially done	lgd_segment_min_cohort	Tests\Feature\Pd\PdAndLgdEngineTest::test_a_loan_closed_at_nil_is_not_a_cure_and_a_nil_start_balance_is_no_member
5.5.17(b), B5.5.44	Time value of money	Partially done		Tests\Feature\Ecl\EclDiscountRateServiceTest::test_an_unlocked_or_missing_eir_is_unresolved_never_defaulted; Tests\Feature\Ecl\EclDiscountRateServiceTest::test_a_floating_facility_is_discounted_at_its_current_eir_not_the_original; Tests\Feature\Ecl\TimePhasedEclFloatingRateTest::test_a_floating_loan_with_no_rate_charged_for_the_month_is_unresolved_never_discounted_at_the_original_eir; Tests\Feature\Ecl\EclGoldenNumberTest::discounted_mode_without_the_time_phased_engine_is_refused_never_proxied; Tests\Feature\Reports\BookedAllowanceAndWriteOffsTest::test_the_basis_of_the_allowance_names_the_loans_without_a_discounted_figure
5.5 WRITE-OFF AND PRESENTATION				

Ref	Section	Status	Governance setting	Test
5.4.4, B5.4.9	Write-off	Evidence needed from MAIIC		Tests\Feature\Eir\WriteOffTest::test_a_closure_credit_in_a_batch_that_debits_the_write_off_gl_is_a_write_off; Tests\Feature\Eir\WriteOffTest::test_a_write_off_batch_in_the_monthly_ledger_feed_is_read; Tests\Feature\Pd\PdSegmentationTest::test_a_nil_start_balance_is_no_observation_and_a_write_off_in_the_window_is_a_default
THE MEGA FARM PROGRAMME				
16	The Mega Farm loans in the allowance	Evidence needed from MAIIC	mega_farms_scope; megafarm_pd_method; megafarm_scalar_ceiling	Tests\Feature\Ecl\MegaFarmProgrammeSeparationTest::test_the_programme_lgd_is_its_own_cohort_and_a_missing_cohort_declines

Baselines - the acceptance ties read from the system

Generated 2026-10-14 10:44:30 from maiic_ifrs9_wire.

Tie	Expected	System figure	Result
Interest posted Jan 2025 to Jul 2026 (ledger, types 303 and 120)	5,293,988,207.06	5,293,988,207.06	PASS
Derived carrying amount against the stored report, in-scope loan GLs (over 2 cents)	0 differences	0 of 2495 account-months	PASS
Mega Farm: account-months whose MF_03 ledger differs from the Loan Book Report (reported, not a test)	reported	77540 account-months	PASS
Year-end interest batch contra on 4215	32,956,675.55	32,956,675.55	PASS
Year-end interest batch contra on 4216	-21,733,262.70	-21,733,262.70	PASS
Loan book 1050101 at 31 Dec 2025 against the trial balance	1,368,737,810.68	1,368,737,810.68	PASS
Loan book 1050102 at 31 Dec 2025 against the trial balance	6,434,045,052.56	6,434,045,052.56	PASS
Loan book 1050201 at 31 Dec 2025 against the trial balance (accepted exception, spec 3.5: -400,000.00)	1,402,393,139.46	1,401,993,139.46	PASS
Loan book 1050202 at 31 Dec 2025 against the trial balance (accepted exception, spec 3.5: 1,000,000.00)	3,808,288,656.15	3,809,288,656.15	PASS
Loan book 1050401 at 31 Dec 2025 against the trial balance	1,827,148,120.69	1,827,148,120.69	PASS

Tie	Expected	System figure	Result
Pack inventory: ebanker-pack-2026-10-09: every file as committed and landed with its manifest rows and SHA-256; 13 received files not landed, each with its reason in the manifest	12 extracts, 111,513 rows	12 extracts, 111,513 rows	PASS
Pack inventory: ebanker-pack-2026-10-12: every file as committed and landed with its manifest rows and SHA-256; 2 received files not landed, each with its reason in the manifest	39 extracts, 511,785 rows	39 extracts, 511,785 rows	PASS
Pack inventory: ebanker-pack-2026-10-12-p5: every file as committed and landed with its manifest rows and SHA-256; 1 received files not landed, each with its reason in the manifest	8 extracts, 470 rows	8 extracts, 470 rows	PASS
Pack inventory: ebanker-pack-2026-10-12-p6: every file as committed and landed with its manifest rows and SHA-256; 0 received files not landed, each with its reason in the manifest	3 extracts, 236 rows	3 extracts, 236 rows	PASS
Pack inventory: ebanker-pack-2026-10-13-p7: every file as committed and landed with its manifest rows and SHA-256; 0 received files not landed, each with its reason in the manifest	6 extracts, 32,244 rows	6 extracts, 32,244 rows	PASS
Pack inventory: ebanker-pack-2026-10-13-p8: every file as committed and landed with its manifest rows and SHA-256; 6 received files not landed, each with its reason in the manifest	4 extracts, 103 rows	4 extracts, 103 rows	PASS
Pack inventory: ebanker-pack-2026-10-14: every file as committed and landed with its manifest rows and SHA-256; 13 received files not landed, each with its reason in the manifest	2 extracts, 426 rows	2 extracts, 426 rows	PASS
Pack inventory: manifest.json (packs 1 and 2, data dictionary, trial balances, take-on): every file as committed and landed with its manifest rows and SHA-256; 3 received files not landed, each with its reason in the manifest; 30 workbooks and scripts, 23 landed by the bootstrap	54 extracts, 85,174 rows, 23 workbooks	54 extracts, 85,174 rows, 23 workbooks	PASS

Tie	Expected	System figure	Result
Take-on worksheet (CV_10) against the take-on postings of 31 Jul 2024: term loans on the worksheet with take-on postings	77.00	77.00	PASS
Take-on worksheet (CV_10) against the take-on postings of 31 Jul 2024: loans whose principal taken on (301) = NEWDISBURS_AMT, to the cent	76.00	76.00	PASS
Take-on worksheet (CV_10) against the take-on postings of 31 Jul 2024: loans whose opening interest (303) = TOTAL_INT_CHARG, to the cent	76.00	76.00	PASS
Take-on worksheet (CV_10) against the take-on postings of 31 Jul 2024: loans whose opening recoveries (305) = TOTA_REPAY_UPTO, to the cent	72.00	72.00	PASS
Ledger copy CUMVOUCH_BK240824 (CV_04a) to 31 Dec 2023 against the GL opening balances of 1 Jan 2024 (CV_14), 4 loan GLs: sum of the differences	0.50	0.50	PASS
Loan book 2025-12, MAIIC's loan GLs: carrying amount	no golden value committed (book_2025-12_maiic_amount)	14,841,212,779.54	FAIL
Loan book 2025-12, MAIIC's loan GLs: rows	no golden value committed (book_2025-12_maiic_loans)	123.00	FAIL
Loan book 2025-12, Mega Farm programme GLs: carrying amount	no golden value committed (book_2025-12_megafarm_amount)	54,498,560,819.60	FAIL
Loan book 2025-12, Mega Farm programme GLs: accounts with a balance	no golden value committed (book_2025-12_megafarm_accounts)	3,671.00	FAIL
Loan book 2026-09, MAIIC's loan GLs: carrying amount	no golden value committed (book_2026-09_maiic_amount)	23,189,580,235.43	FAIL

Tie	Expected	System figure	Result
Loan book 2026-09, MAIIC's loan GLs: rows	no golden value committed (book_2026-09_maiic_loans)	134.00	FAIL
Loan book 2026-09, Mega Farm programme GLs: carrying amount	no golden value committed (book_2026-09_megafarm_amount)	56,180,761,113.43	FAIL
Loan book 2026-09, Mega Farm programme GLs: accounts with a balance	no golden value committed (book_2026-09_megafarm_accounts)	3,618.00	FAIL
Ledger write-offs on MAIIC loans (WriteOffClassifier): postings	no golden value committed (ledger_writeoffs_postings)	22.00	FAIL
Ledger write-offs on MAIIC loans (WriteOffClassifier): amount	no golden value committed (ledger_writeoffs_amount)	60,778,466.86	FAIL
ECL 2025-12, MAIIC's book: undiscounted	no golden value committed (ecl_2025-12_maiic_undiscounted)	3,955,543,846.37	FAIL
ECL 2025-12, MAIIC's book: discounted at the EIR (booked)	no golden value committed (ecl_2025-12_maiic_discounted)	2,593,174,242.30	FAIL
ECL 2025-12, MAIIC's book: loans discounted at the EIR	no golden value committed (ecl_2025-12_maiic_discounted_loans)	56.00	FAIL
ECL 2025-12, MAIIC's book, stage 1: discounted	no golden value committed (ecl_2025-12_maiic_stage1_discounted)	61,318,291.61	FAIL
ECL 2025-12, MAIIC's book, stage 2: discounted	no golden value committed (ecl_2025-12_maiic_stage2_discounted)	315,409,106.41	FAIL

Tie	Expected	System figure	Result
ECL 2025-12, MAIIC's book, stage 3: discounted	no golden value committed (ecl_2025-12_maiic_stage3_discounted)	2,216,446,844.28	FAIL
ECL 2025-12, Mega Farm programme (latest run)	no golden value committed (ecl_2025-12_megafarm_programme)	none	FAIL
ECL 2025-12, Mega Farm: MAIIC's share	no golden value committed (ecl_2025-12_megafarm_maiic_share)	none	FAIL
ECL 2026-09, MAIIC's book: undiscounted	no golden value committed (ecl_2026-09_maiic_undiscounted)	11,569,458,915.88	FAIL
ECL 2026-09, MAIIC's book: discounted at the EIR (booked)	no golden value committed (ecl_2026-09_maiic_discounted)	7,118,643,553.12	FAIL
ECL 2026-09, MAIIC's book: loans discounted at the EIR	no golden value committed (ecl_2026-09_maiic_discounted_loans)	62.00	FAIL
ECL 2026-09, MAIIC's book, stage 1: discounted	no golden value committed (ecl_2026-09_maiic_stage1_discounted)	219,546,033.90	FAIL
ECL 2026-09, MAIIC's book, stage 2: discounted	no golden value committed (ecl_2026-09_maiic_stage2_discounted)	183,786,390.69	FAIL
ECL 2026-09, MAIIC's book, stage 3: discounted	no golden value committed (ecl_2026-09_maiic_stage3_discounted)	6,715,311,128.53	FAIL
ECL 2026-09, Mega Farm programme (latest run)	no golden value committed (ecl_2026-09_megafarm_programme)	45,782,301,503.37	FAIL

Tie	Expected	System figure	Result
ECL 2026-09, Mega Farm: MAIIC's share	no golden value committed (ecl_2026-09_megafarm_maiic_share)	2,289,115,075.17	FAIL
EIR contracts locked	no golden value committed (eir_contracts_locked)	14.00	FAIL
EIR contracts blocked	no golden value committed (eir_contracts_blocked)	33.00	FAIL
EIR contracts solved (an EIR computed)	no golden value committed (eir_contracts_solved)	14.00	FAIL
EIR contracts in the contract master	no golden value committed (eir_contracts)	144.00	FAIL
Revenue 2024: EIR interest recognised	no golden value committed (revenue_2024_eir_interest)	25,532,678.70	FAIL
Revenue 2024: cash received	no golden value committed (revenue_2024_cash_received)	34,200,000.00	FAIL
Revenue 2024: roll-forward rows	no golden value committed (revenue_2024_rows)	20.00	FAIL
Revenue 2025: EIR interest recognised	no golden value committed (revenue_2025_eir_interest)	663,970,866.88	FAIL
Revenue 2025: cash received	no golden value committed (revenue_2025_cash_received)	1,158,267,060.20	FAIL
Revenue 2025: roll-forward rows	no golden value committed (revenue_2025_rows)	126.00	FAIL

Tie	Expected	System figure	Result
Revenue 2026: EIR interest recognised	no golden value committed (revenue_2026_eir_interest)	544,517,024.26	FAIL
Revenue 2026: cash received	no golden value committed (revenue_2026_cash_received)	2,381,610,987.63	FAIL
Revenue 2026: roll-forward rows	no golden value committed (revenue_2026_rows)	113.00	FAIL
RBM return 2026-09: facilities classified loss (paragraph 12 list)	no golden value committed (rbm_2026-09_loss_facilities)	10.00	FAIL
RBM return 2026-09: balance of the loss list	no golden value committed (rbm_2026-09_loss_balance)	1,510,956,083.45	FAIL

Findings - items that need a decision or a fix

F1. The PD, LGD and ECL are the bootstrap's, not MAIIC's (5.5.1, 5.5.17) - Open

- **What was found:** The PD and LGD engines run under the bootstrap label; no MAIIC reviewer has approved a segment run, a fit or the scenario set. On the fresh bootstrap the main-book ECL at September 2026 is 11.57bn undiscounted (7.12bn discounted on 62 loans, 72 unresolved). (Until 14 October 2026 this finding quoted 8.33bn, which the Contract Schedule 1 workbook gave as 8.94bn; neither is current.)
- **Impact:** The ECL is a system figure awaiting review.
- **Recommended action:** A reviewer approves the segment runs, a fit and the set; the month is re-run discounted and signed.
- **Owner:** Dr Thom / the reviewer

F2. No fit is approved by a MAIIC reviewer (B5.5.49) - Open

- **What was found:** On the fresh bootstrap the guardrail applied 32 of 209 fits and declined 177; none is approved for the route.
- **Impact:** The forward-looking adjustment is the overlay at zero until then.
- **Recommended action:** A reviewer approves a fit on the Regression (FLI adjustments) screen; a second person reviews.
- **Owner:** Dr Thom / the reviewer

F3. The Mega Farm provision basis is undocumented (16) - Open

- **What was found:** The 2025 statements carry K39.76bn against K48.7bn Stage 3. On the fresh bootstrap (staging before 216aee2) the December 2025 programme book of 3,692 loans (54.50bn) is 3,219 loans Stage 1 (47.25bn) and 471 Stage 3 (7.25bn).
- **Impact:** The ECL golden number at 31 December 2025 cannot be set.
- **Recommended action:** The CFO confirms D30 and the provisioning method; the next bootstrap on 216aee2 restages the programme on the chart in force.

- **Owner:** Dr Thom / Dupleix

F4. The reports carry the undiscounted allowance (5.5.17(b)) - In progress

- **What was found:** The discounted allowance is held in ecl_value_discounted; the IFRS 9 note, ECL reconciliation, RBM Return and other reports read ecl_value. Status 14 October 2026: the reports carry the booked allowance with the undiscounted measure as a reconciling line, and name the loans with no discounted figure (38 at September 2026); the note fails closed until there are none.
- **Impact:** The disclosed allowance is not wholly discounted until those loans carry a locked EIR.
- **Recommended action:** Lock the EIRs of the named loans (fee reviews), then run the month-end ECL again.
- **Owner:** Dupleix / MAIIC reviewers

F5. Collateral is not in the LGD (B5.5.55) - Open

- **What was found:** The LGD engine reads no collateral; the collateral register is empty on the fresh bootstrap.
- **Impact:** The cash flows from collateral are not reflected in the shortfall.
- **Recommended action:** Credit supplies the collateral and valuations; Dupleix adds the collateral recovery to the LGD.
- **Owner:** Credit / Dupleix

Section-by-section audit

5.5 RECOGNITION OF EXPECTED CREDIT LOSSES

5.5.1. A loss allowance for expected credit losses - *Done*

- **What it requires:** Recognise a loss allowance for expected credit losses on a financial asset measured at amortised cost.
- **What the engine does:** ifrs9:segment-ecl computes the allowance per loan as EAD x PD x LGD on the post-FLI PD for every portfolio of a month and fails closed, naming the loans, when a stage, PD or LGD is missing or a portfolio mixes Mega Farm with MAIIC's own loans; the bootstrap and the month-end (fli:month-end) run it for every period. On the fresh bootstrap the main book carries an allowance in every month (11.57bn undiscounted at September 2026).
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said the PD engine must run for a figure.
- **Compliance comment:** Compliant; the figures are system figures under the bootstrap label, not MAIIC-approved (finding F1).
- **Where to see it:** ECL Processing, ECL Calculation: /expected-credit-loss/list
- **Test that proves it:** `Tests\Feature\Ecl\EclGoldenNumberTest::it_computes_golden_ecl_per_stage_for_prefli_collection_lgd; Tests\Feature\Ecl\TimePhasedEclServiceTest`
- **Reviewer sign-off:** _____ Date: _____

5.5.3. Lifetime ECL on a significant increase in credit risk - *Done*

- **What it requires:** At each reporting date, measure the loss allowance at lifetime ECL if the credit risk has increased significantly since initial recognition.
- **What the engine does:** StagingService stages every period: Stage 2 from 31 days past due in every class, the SICR flag from the qualitative groups; the ECL takes the lifetime PD for Stage 2, compounding beyond the 12-month PD. Since 216aee2 the days past due read E-Banker's instalment chart in force at each month-end.
- **General comment:** Spec v4 section 3.6, decision D31.
- **Compliance comment:** Compliant.
- **Where to see it:** Settings, Staging basis: /setting?tab=staging; IFRS 9 Model Setup, Staging & SICR Rules: /sicr-groups; ECL Processing, ECL Calculation: /expected-credit-loss/list
- **Governance setting:** `dpd_basis`
- **Test that proves it:** `Tests\Feature\Ecl\TimePhasedEclServiceTest::test_a_stage_two_lifetime_compounds_beyond_the_twelve_month_pd; Tests\Feature\Eir\StagingServiceTest::test_the_trigger_re`

`ads_the_chart_in_force_at_the_month_end_not_a_later_one`

- **Reviewer sign-off:** _____ Date: _____

5.5.5. 12-month ECL where credit risk has not increased significantly - *Done*

- **What it requires:** Measure the loss allowance at 12-month ECL where the credit risk has not increased significantly since initial recognition.
- **What the engine does:** Stage 1 loans take the 12-month PD window; the staging writes the stage every period.
- **Compliance comment:** Compliant.
- **Where to see it:** ECL Processing, ECL Calculation: /expected-credit-loss/list
- **Test that proves it:** `Tests\Feature\Ecl\TimePhasedEclServiceTest::test_stage_one_builds_a_twelve_month_marginal_pd_curve_and_discounts_each_shortfall`
- **Reviewer sign-off:** _____ Date: _____

5.5.9 to 5.5.11. Determining a significant increase in credit risk; the 30-days-past-due rebuttable presumption - *Done*

- **What it requires:** Assess the change in the risk of default since initial recognition using reasonable and supportable information; there is a rebuttable presumption that credit risk has increased significantly when payments are more than 30 days past due.
- **What the engine does:** Stage 2 from 31 days past due in every facility class, counted from the oldest overdue instalment, with the bucket ageing beside it. The rebuttal is recorded on the threshold rows (staging_thresholds.rebuttal_basis); the LONG_TERM row proposing Stage 2 at 91 days for medium- and long-term facilities only is seeded future-dated (2099) and inactive pending CFO sign-off. Since 14 October 2026 a threshold row is proposed by one person and approved by another on Settings, Staging basis, forward only and never into a locked month, and staging reads only approved rows, at the period end, with no 31/181 fallback in the legacy importer (audit H11). The staging_rebuttal catalogue setting is marked recorded, not yet applied: the rebuttal is the threshold row with its evidence.
- **General comment:** The O13 rebuttal paper of 8 October 2026. Until 14 October 2026 the threshold rows were seeded and read-only on screen.
- **Compliance comment:** Compliant; the presumption is applied and the rebuttal awaits the CFO.
- **Where to see it:** Settings, Staging basis: /setting?tab=staging; IFRS 9 Model Setup, Staging & SICR Rules: /sicr-groups; Governance: Governance Centre /eir-governance
- **Governance setting:** `dpd_basis; staging_rebuttal (recorded, not yet applied)`
- **Test that proves it:** `Tests\Feature\Eir\StagingThresholdSeederTest::test_the_long_term_stage_2_rebuttal_is_inactive_until_signed; Tests\Feature\Eir\GoLiveControlsTest::test_a_threshold_is_proposed_by_one_person_and_approved_by_another_before_staging_reads_it; Tests\Feature\Eir\StagingClassifierTest::test_with_no_rule_in_force_the_classifier_stops_and_names_it; Tests\Feature\Eir\StagingServiceTest`
- **Reviewer sign-off:** _____ Date: _____

B5.5.37. Definition of default; the 90-days-past-due rebuttable presumption - *Done*

- **What it requires:** Apply a definition of default consistent with internal credit risk management; default occurs no later than 90 days past due unless reasonable and supportable information supports a more lagging criterion.
- **What the engine does:** Stage 3 from 91 days for a facility of 12 months or less and for the Mega Farm class, and from 181 days for a medium- or long-term facility, as the RBM directive classifies them; the four-consecutive-missed-instalments trigger lifts a loan to Stage 3 beside the day count, reading the chart in force and net cash (216aee2). The more lagging criterion rests on the directive, cited on each threshold row.
- **General comment:** Spec v4 section 3.6 compares the policy (90 days), the November 2025 model (181) and the directive (181/91).
- **Compliance comment:** Compliant; the rebuttal is documented on the threshold rows.

- **Where to see it:** Settings, Staging basis: /setting?tab=staging; IFRS 9 Model Setup, Staging & SICR Rules: /sicr-groups
- **Governance setting:** `dpd_basis; stage3_missed_instalments`
- **Test that proves it:** `Tests\Feature\Eir\StagingServiceTest::test_the_directive_thresholds_by_tenor_class_and_the_bucket_fallback; Tests\Feature\Eir\StagingThresholdSeederTest::test_medium_and_long_term_facilities_are_stage_3_at_181_days_with_the_rebuttal_cited`
- **Reviewer sign-off:** _____ Date: _____

5.5 MEASUREMENT

5.5.17. Measurement of expected credit losses - *Partially done*

- **What it requires:** ECL is an unbiased, probability-weighted amount determined by evaluating a range of possible outcomes, the time value of money, and reasonable and supportable information about past events, current conditions and forecasts.
- **What the engine does:** The PD and LGD by segment, the forward-looking route once per scenario and the weighted ECL run for every period; discounting at the EIR is the month-end default since 216aee2. What is missing: no fit is approved by a MAIIC reviewer, so the route holds the PD (overlay at zero); the governed scenario set is approved only under the bootstrap label; the legacy synthetic scenario assumptions are still seeded APPROVED (EclScenarioAssumptionSeeder); and the reports read the undiscounted ecl_value (5.5.17(b) row).
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said the PD engine had not run on the clean install.
- **Compliance comment:** Partially done: the mechanics are in place; an approved fit and set, and the discounted figure in the reports, are awaited.
- **Where to see it:** ECL Processing, ECL Calculation: /expected-credit-loss/list; Forward-Looking Information, Scenario Sets: /scenario-sets
- **Governance setting:** `scenario_weighting_method; fli_adjustment_route`
- **Test that proves it:** `Tests\Feature\FLI\FliRouteServiceTest::test_the_ecl_on_the_post_fli_pd_is_the_weighted_ecl_across_scenarios; Tests\Feature\Ecl\TimePhasedEclServiceTest::test_scenario_weights_combine_into_one_allowance`
- **Reviewer sign-off:** _____ Date: _____

B5.5.41 to B5.5.43. Probability-weighted outcomes - *Done*

- **What it requires:** The estimate reflects the possibility that a credit loss occurs and the possibility that none occurs; a range of scenarios, at least two, weighted by their probabilities.
- **What the engine does:** The governed scenario set: three or more scenarios, weights summing to 100, the base at least 40 percent and no weight above 60, every scenario but the base a shock on the base path; the first set (Base 50, Upside 15, Downside 25, Severe 10) is seeded as a proposal for the CFO. On the fresh bootstrap one set (September 2026) is APPROVED under the bootstrap label.
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2).
- **Compliance comment:** Compliant.
- **Where to see it:** Forward-Looking Information, Scenario Sets: /scenario-sets
- **Governance setting:** `scenario_minimum_count; scenario_weight_bounds; scenario_calibration_note`
- **Test that proves it:** `Tests\Feature\Scenario\ScenarioSetServiceTest::test_a_set_that_breaks_a_rule_is_refused_with_the_reason`
- **Reviewer sign-off:** _____ Date: _____

B5.5.49 to B5.5.54. Reasonable and supportable information; forward-looking information - *Done*

- **What it requires:** Use information reasonably available without undue cost or effort, including forecasts of future economic conditions; a statistical relationship used must be supportable.
- **What the engine does:** Malawi macro series with their provenance (13 variables, 261 data points on the fresh bootstrap; none yet carries a release date, so revisions replace the earlier value); the correlation finder sweeps

every driver against every credit-loss proxy over the lag grid under the guardrail (expected sign, R2 cutoff, p-value, minimum observations) and declines a failing fit with the reason. On the fresh bootstrap: 209 fits, 32 applied by the guardrail, 177 declined, none approved.

- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said 39 applied and 170 declined, from the 8 October install.
- **Compliance comment:** Compliant; an applied fit is a proposal under maker-checker.
- **Where to see it:** Forward-Looking Information: Overview /fli; Correlation Finder /fli-correlation; Regression /fli-adjustments; transmission methods /fli/transmission-methods; Macro Statistics: /macro-statistics
- **Governance setting:** `fli_expected_sign_test; fli_r2_cutoff; fli_min_observations; fli_alpha; macro_source_precedence`
- **Test that proves it:** `Tests\Feature\FLI\FliBridgeServiceTest; Tests\Feature\Macro\MacroImportPrecedenceTest`
- **Reviewer sign-off:** _____ Date: _____

B5.5.52 (transmission). How forward-looking information reaches the PD - *Done*

- **What it requires:** The method of adjusting the PD for the forecast must be documented and supportable.
- **What the engine does:** Governed transmission methods, each with a card (what it does, the formula, its preconditions checked live against the data, a worked example on one loan); a method whose preconditions are not met cannot be selected; the scalar is seeded because the history cannot yet support the richer methods.
- **General comment:** Spec v4 section 14.7.
- **Compliance comment:** Compliant.
- **Where to see it:** Forward-Looking Information: Overview /fli; Correlation Finder /fli-correlation; Regression /fli-adjustments; transmission methods /fli/transmission-methods; Governance: Governance Centre /eir-governance
- **Governance setting:** `fli_transmission_method; fli_asset_correlation`
- **Test that proves it:** `Tests\Feature\FLI\TransmissionMethodCatalogueTest`
- **Reviewer sign-off:** _____ Date: _____

B5.5.1 to B5.5.6, B5.5.28 to B5.5.30. Collective assessment and segmentation - *Done*

- **What it requires:** Where information is not available at the instrument level, assess on a collective basis grouped by shared credit-risk characteristics.
- **What the engine does:** PdSegmentationService and LgdSegmentationService measure the PD and LGD per portfolio on the governed basis, a thin segment taking its parent's or the pooled figure under the governed rule, with the lineage on each loan; the Mega Farm programme is its own segment, kept out of MAIIC's PD, LGD and ECL, and a portfolio mixing the two is refused (216aee2). Both run for every period in the bootstrap and the month-end.
- **General comment:** Until 14 October 2026 this row was Partially done and said the transition matrices were not in the bootstrap chain.
- **Compliance comment:** Compliant.
- **Where to see it:** IFRS 9 Model Setup, PD Model and LGD Model; Reports: PD by portfolio /ifrs9-reports/sector-ecl
- **Governance setting:** `pd_segmentation_basis; pd_segment_min_observations; pd_segment_thin_rule; lgd_segment_min_cohort`
- **Test that proves it:** `Tests\Feature\Pd\PdSegmentationTest::test_each_portfolio_gets_its_own_pd_or_its_parents_with_the_lineage_recorded; Tests\Feature\Ecl\MegaFarmProgrammeSeparationTest::test_the_programme_portfolio_is_kept_out_of_the_maiic_scope_and_a_blended_book_is_refused`
- **Reviewer sign-off:** _____ Date: _____

B5.5.55. Loss given default and collateral - *Partially done*

- **What it requires:** The cash shortfalls reflect the cash flows expected from collateral and other credit enhancements that are integral to the contractual terms.
- **What the engine does:** The LGD is the Stage 3 cohort workout per portfolio (cure, recovery discounted, write-off as loss; a written-off loan left at nil is no longer a cure since 216aee2). Collateral does not enter it: the engine reads no collateral, and on the fresh bootstrap collateral_registers and collateral_allocations are empty although the E-Banker security register (P2_11, 185 rows) is landed.
- **General comment:** Until 14 October 2026 this row was referenced B5.5.33 to B5.5.35 and marked Evidence needed; the collateral link is also Dupleix's to build.
- **Compliance comment:** Partially done: Credit supplies the collateral and valuations; Dupleix brings collateral into the LGD.
- **Where to see it:** IFRS 9 Model Setup, LGD Model; Collateral Management, Collateral Register: /collateral/register; Reports: LGD and collateral /ifrs9-reports/lgd-collateral
- **Governance setting:** `lgd_segment_min_cohort`
- **Test that proves it:** `Tests\Feature\Pd\PdAndLgdEngineTest::test_a_loan_closed_at_nil_is_not_a_cure_and_a_nil_start_balance_is_no_member`
- **Reviewer sign-off:** _____ Date: _____

5.5.17(b), B5.5.44. Time value of money - *Partially done*

- **What it requires:** Discount the expected credit losses at the EIR determined at initial recognition (or an approximation).
- **What the engine does:** TimePhasedEclService discounts each loan's expected shortfalls at its locked original EIR and a floating-rate loan's at its current EIR for the month (EIR_CURRENT_FLOATING: the rate the revenue run accrued at, or the same D38 re-estimation from the rate charged; B5.5.44); an unlocked or missing EIR, or a floating loan with no current EIR, is unresolved with the reason, never defaulted or given the original rate; a Stage 3 loss at the reporting date needs no rate. Since 14 October 2026 EclDiscountRateService (the LGD recovery discounting) applies the same rule, so no basis is labelled EIR_ORIGINAL_FLOATING_PROXY any more. Since 216aee2 the month-end and the bootstrap run discounted by default: ecl_value keeps the engine's measure and ecl_value_discounted holds the discounted allowance, shown on the ECL screen. Since 14 October 2026 every report carries the booked allowance (BookedEcl: discounted at the original EIR; the Mega Farm programme measure; a loan with no discounted figure at its undiscounted ECL, counted and named), with the reconciliation from the reporting-date measure (the lifetime projection and the effect of discounting); the IFRS 9 note fails closed while any loan lacks its discounted allowance. On the fresh bootstrap at September 2026, 62 main-book loans are discounted (7,118,643,553.12) and 38 with an ECL have no discounted figure (4,460,558,036.40 undiscounted; most without a locked EIR).
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said 'the contractual rate where none is locked', which the code never did; and the reports read the undiscounted ecl_value.
- **Compliance comment:** Partially done: the reports carry the discounted allowance; 38 loans still need a locked EIR before the allowance is wholly discounted.
- **Where to see it:** ECL Processing, ECL Calculation: /expected-credit-loss/list
- **Test that proves it:** `Tests\Feature\Ecl\EclDiscountRateServiceTest::test_an_unlocked_or_missing_eir_is_unresolved_never_defaulted; Tests\Feature\Ecl\EclDiscountRateServiceTest::test_a_floating_facility_is_discounted_at_its_current_eir_not_the_original; Tests\Feature\Ecl\TimePhasedEclFloatingRateTest::test_a_floating_loan_with_no_rate_charged_for_the_month_is_unresolved_never_discounted_at_the_original_eir; Tests\Feature\Ecl\EclGoldenNumberTest::discounted_mode_without_the_time_phased_engine_is_refused_never_proxied; Tests\Feature\Reports\BookedAllowanceAndWriteOffsTest::test_the_basis_of_the_allowance_names_the_loans_without_a_discounted_figure`
- **Reviewer sign-off:** _____ Date: _____

5.5 WRITE-OFF AND PRESENTATION

5.4.4, B5.4.9. Write-off - *Evidence needed from MAIIC*

- **What it requires:** Write off the gross carrying amount when there is no reasonable expectation of recovery.
- **What the engine does:** WriteOffClassifier (220a5a6) identifies ledger write-offs (type 300; a batch debiting GL 1064; ACCOUNT CLOSURE on a status-H account) for the loan-book build, the contract inputs and the EIR cash; the PD cohort counts a write-off as a default (216aee2); the register records a write-off not on the loan account under two-person approval; since 14 October 2026 the same source (WriteOffSource) feeds the IFRS 9 note, the reconciliations and the RBM Return, and the monthly feed M09_01 is read for write-off batch legs. On the fresh bootstrap: 22 ledger write-offs of 60,778,466.86 (20 of them, 38,492,032.54, between December 2025 and September 2026); the register is empty.
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said no write-off had been posted since the take-on.
- **Compliance comment:** Evidence needed: MAIIC's write-off policy and Finance's confirmation of status H.
- **Where to see it:** EIR Data, Write-offs: /eir-write-offs
- **Test that proves it:** `Tests\Feature\Eir\WriteOffTest::test_a_closure_credit_in_a_batch_that_debits_the_write_off_gl_is_a_write_off; Tests\Feature\Eir\WriteOffTest::test_a_write_off_batch_in_the_monthly_ledger_feed_is_read; Tests\Feature\Pd\PdSegmentationTest::test_a_nil_start_balance_is_no_observation_and_a_write_off_in_the_window_is_a_default`
- **Reviewer sign-off:** _____ Date: _____

THE MEGA FARM PROGRAMME

16. The Mega Farm loans in the allowance - *Evidence needed from MAIIC*

- **What it requires:** Loans extended under the Government's Mega Farm programme (losses to the fund; MAIIC's 5 percent share) must be in the ECL on a documented basis.
- **What the engine does:** The programme's monthly books (MF_01 and MF_03) are landed: on the fresh bootstrap 3,692 loans on six of the programme's seven GLs (1050301 to 1050305 and 1050307) with 56.18bn gross at September 2026, 3,559 of them over 90 days past due. Since 216aee2 the programme is segmented as its own portfolio, kept out of MAIIC's PD, LGD and ECL, and provided by megafarm:ecl from its own LGD cohort under the governed PD method; MAIIC's share is read from the mega_farms_scope text (it falls back to 1.0 when the text names no share). Not built: the interest split, the maize receivable and the fund liability of spec 16.4 and 16.5 (they wait on D30).
- **General comment:** Fresh bootstrap maiic_ifrs9_results2 (13 October 2026, before commit 216aee2). Until 14 October 2026 the row said 3,490 loans (K54.85bn) from a November 2025 book.
- **Compliance comment:** Evidence needed: the CFO's confirmation of D30; how the K39.76bn provision of 2025 was arrived at.
- **Where to see it:** ECL Processing, Mega Farm Programme: /megafarm; Governance: Governance Centre /eir-governance
- **Governance setting:** `mega_farms_scope; megafarm_pd_method; megafarm_scalar_ceiling`
- **Test that proves it:** `Tests\Feature\Ecl\MegaFarmProgrammeSeparationTest::test_the_programme_lgd_is_its_own_cohort_and_a_missing_cohort_declines`
- **Reviewer sign-off:** _____ Date: _____